MILITARY SPECIFICATION SHEET

CABLE ASSEMBLY, AIRCRAFT AUDIO, CABLE (RETRACTILE)

This specification sheet is approved for use by the Naval Air Systems Command, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of MIL-C-22442(AS).

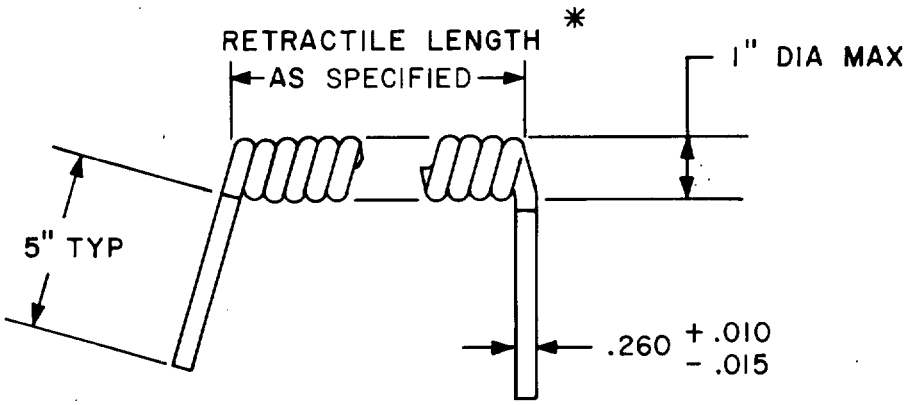


FIGURE 1
P/N M22442/3

* ±1" PER RETRACTED FOOT OR FRACTION THEREOF

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FSC 5995

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APPLICABLE DOCUMENTS:

MIL-C-572	Cords, Yarns and Monofilaments Organic Synthetic Fiber
MIL-I-3930	Insulating and Jacketing Compounds, Electrical (for Cables, Cords and Wires)
MIL-C-11997	Cord Assembly, Electrical CX-2151( )/U (Retractile)
MIL-C-13273	Cord, Electrical (Retractile 2, 3 and 4 Conductor WD-9/U, WT-2/U and WF-4/U)
MIL-STD-104	Limit for Electrical Insulation Color

REQUIREMENTS:

1. All performance requirements and construction requirements except as modified by this specification sheet shall be in accordance with MIL-C-13273.

2. Conductors. Each conductor shall be equivalent to 404 circular mils (No. 24 AWG) and shall consist of 41 strands of No. 40 AWG tinned soft cadmium copper alloy, having a 1 percent nominal content of cadmium. It shall be tight bunch stranded with a lay not exceeding 0.75 inch. Prior to stranding, the elongation of individual tinned soft cadmium copper alloy strands shall be not less than 10 percent and the tensile strength shall not exceed 55,000 pounds per square inch. Subsequent to stranding, and prior to insulating, the elongation of individual strands shall not be less than 5 percent and the tensile strength shall not exceed 58,000 pounds per square inch. Each conductor in the finished cord shall have a nominal DC resistance of 39.0 ohms per 1000 feet at or corrected to a temperature of 20°C.

3. Insulation. The insulation for the retractile cords shall be Type IR (or Type IH) rubber compound conforming to the requirements of MIL-I-3930 and capable of meeting the requirements specified herein.

4. Dimensions of insulated conductors. The diameter of the insulated stranded conductors shall be 0.057 inch nominal and shall not exceed 0.063 inch. The minimum wall thickness shall not be less than 0.009 inch.

5. Colors. Colors shall conform to MIL-STD-104. The color of the insulation on the 5 conductors shall be black, white, red, green, and orange.

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6. Separators. Separators shall consist of a serving, a knitted barrier, or a close braid of synthetic yarn conforming to Type P, AR or SAR of MIL-C-572 or of soft cotton. If, however, the insulating compound applied is free-stripping, the separator may be omitted.

7. Purity of yarns. The purity of yarns used as a separator under insulation shall be determined in accordance with MIL-C-11997.

8. Staycord. Staycords shall consist of synthetic textile yarn conforming to Type P, AR or SAR of MIL-C-572. The cord shall have a minimum breaking strength of 25 pounds, and shall not abrade or otherwise damage the insulation when the cord is bent or stretched.

9. Binder. A binder if used shall consist of a suitable number of ends of synthetic textile yarns conforming to Type P, AR or SAR of MIL-C-572, or of soft cotton.

10. Shield. The shield shall consist of 32 nominal strands of No. 36 AWG tinned soft cadmium copper alloy having a 1 percent nominal content of cadmium and served over an insulated conductor in an even helix (laid adjacent to each other) with a nominal 90% coverage. A served shield shall be applied over each black, red, and green insulated conductor separately.

11. Outer jacket. The cabled conductors shall be jacketed with a type JS-L(SBR), JH(EPDM) compound per MIL-I-3930 or a SBR natural rubber blend compound conforming to the performance requirements of MIL-C-13273. The amount of natural rubber in the jacket compound shall not exceed 35 percent by weight of the total SBR and natural rubber in the compound. The amount of sunproofing wax included in either compound shall be in accordance with MIL-I-3930.

12. Jacketed cord. After vulcanization of the jacket the average overall diameter for the 5 conductor cord (with a individual shield around three conductors) shall be .260 $^{+.010}_{-.015}$ inch. The average diameter shall be taken as a mean of the maximum and minimum diameter of 10 measurements at various points along the cord. Diameters of retractile cords shall be taken along the straight portions of the cord at both ends. The jacket thickness at any cross section of the cord shall be .010 minimum. The jacket shall be free-stripping from the insulation. Adapter separator may be used under the jacket.

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13. Construction. The cable shall be constructed of 5 insulated conductors, 3 of which are individually shielded, and a staycord. Filler will be added if necessary to make a circular cross section. The insulated conductors shall be twisted with or around the staycord, and the lay of the conductors shall be 1.75 inch maximum. A suitable binder or separator may be applied over the cabled conductors. The outer jacket shall be Type JS-L-(SBR), JH(EPDM) or SBR natural rubber blend, in accordance with MIL-I-3930, colored black. The minimum jacket thickness at any cross section of the retractile portion of a retractile cord shall not be less than 0.010 inch.

14. Retractile. The jacketed electrical cord shall be formed in a helix with close adjacent turns. The helical cord shall have a straight length of cord on each end tangent to the helical section. The finished electrical cord shall have a left hand lay in its helical section and shall have a smooth and uniform jacket. The helix shall have an overall diameter of not more than 1.0 inches.

15. Quality assurance provision. Sampling inspection and test procedures for the cord construction shall be as outlined in MIL-C-13273.

Preparing Activity:

Navy - AS

Project No. 5995-N107-3

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-C-22442/3(AS) CABLE ASSEMBLY, AIRCRAFT AUDIO, CABLE (RETRACTILE)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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Replaces edition of 1 Jan 72 which may be used.

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